

MD ROBIUL ISLAM

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About Me

I am a Ph.D. student in Electrical and Computer Engineering (ECE) at the University of South Florida (USF), working on AI/ML-driven optimization for satellite and non-terrestrial networks (NTNs) in 5G/6G systems. Investigating multi-layer and multi-orbit NTNs in 6G, with a focus on spectrum management, interference control, and network efficiency. Building on this technical foundation, I am seeking a Ph.D. internship in Wireless Systems Research, 5G/6G R&D, AI/ML for Telecommunications, Network Optimization, RF/Signal Processing Research or Satellite/NTN Engineering to apply my research to real-world, industry-scale problems.

EDUCATION

PhD in Electrical Engineering (ECE Dept.)

Aug 2026 – Present

University of South Florida (USF), Tampa, Florida

- **Research Focus:** AI/ML-driven optimization of multi-layer, multi-orbit Non-Terrestrial Networks (NTNs) for 6G satellite communication, addressing spectrum allocation and sharing, interference management, throughput and spectral efficiency, energy efficiency, and reliability in integrated satellite-terrestrial systems
- **Taken Courses:** Applied Optimization, Linear and Matrix Algebra

MS in Computer and Information Systems Engineering (ECE Dept.)

Sep 2024 – Jul 2026

Tennessee State University (TSU), Nashville, TN | GPA: 3.50/4.00

- **Thesis:** Optimization of 5G Network Slicing via Deep Reinforcement Learning: Comparative Study of PPO, DDPG, and SAC Algorithms
- **Key Courses:** Intelligent Systems & ML, Cyber-Physical Systems Modeling & Simulation, Computer Communication & Networking, Modern Signal Processing, Software Systems Design, Probability & Risk Analysis, Systems Engineering

BSc in Electrical and Electronic Engineering

May 2014 – Nov 2019

Pabna University of Science and Technology (PUST), Bangladesh

- **Thesis:** Fault Analysis of Transmission Lines for Power Systems Using MATLAB Simulation — achieved 3–5% reduction in fault voltage and fault current, demonstrating effective protection system design
- **Key Courses:** Electrical Circuits, Power systems engineering, Electrical power transmission and distribution, Artificial Intelligence, Digital Image Processing, Digital Signal Processing, Network Communication Engineering, Digital Electronics

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug 2026 – Present

Dept. of Electrical and Computer Engineering, University of South Florida | Advisor: Hayder Al-Hraishawi

PhD Project: AI-driven wireless communication

Graduate Research Assistant

Aug 2024 – Jul 2026

Dept. of Electrical and Computer Engineering, Tennessee State University | Advisor: Dr. Sagnika Ghosh

- Designed and implemented a deep RL-based 5G network slicing optimization framework comparing PPO, DDPG, and SAC algorithms for dynamic resource allocation in household IoT environments
- Constructed MATLAB-simulated 5G dataset with SNR, Path Loss, and BER parameters; PPO outperformed DDPG and SAC in network reliability and convergence speed
- Quantified 10–15% improvement in network efficiency under dynamic conditions; manuscript submitted and under peer review
- Applied multi-objective optimization and adaptive RL policy design, developing transferable expertise in intelligent decision-making under constrained, real-world conditions

Undergraduate Research Assistant

2018 – 2019

Dept. of Electrical and Electronic Engineering, PUST, Bangladesh | Advisor: Dr. Md Feroz Ali

- Developed MATLAB Simulink fault analysis model for power transmission lines; achieved 3–5% reduction in fault voltage and fault current
- Simulated fault scenarios across multiple network topologies to evaluate system resilience and protection relay performance

RESEARCH PUBLICATIONS

Journal Articles

- S. Debnath, Md. Buiya, Md. Reza, Md Robiul Islam, et al., “AI-Driven Cybersecurity for Renewable Energy Systems: Detecting Anomalies with Energy-Integrated Defense Data,” *Int'l J. of Applied Mathematics*, vol. 38, no. 5s, pp. 1002–1027, Aug. 2025. DOI: <https://doi.org/10.12732/ijam.v38i5s.367>
- S. Debnath, Md Robiul Islam, et al., “Hybrid Multi-Scale Deep Learning Enhanced Electricity Load Forecasting Using Attention-Based CNN-LSTM Model,” *IEEE Access*, 2025. DOI: <https://doi.org/10.1109/ACCESS.2026.3656545>

Peer-Reviewed Conference Papers

- S. Debnath, Md Robiul Islam, et al., “AI-Driven Hybrid Deep Learning Framework for Short-Term Renewable Energy Forecasting under Extreme Weather Events,” *IEEE ICECIE*, 2025. DOI: <https://doi.org/10.1109/ICECIE66637.2025.11363804>
- S. Debnath, U. Basu, Md Robiul Islam, et al., “Extreme Weather Grid Load Forecasting using Weather-Informed LSTM and Transformer Models,” *IEEE NAPS*, 2025. DOI: <https://doi.org/10.1109/NAPS66256.2025.11272315>
- J. Hassan, Md Robiul Islam, et al., “Comparative Analysis of ML Models for Multi-Class Activity Recognition Using EEG and Motion Sensor Data,” *IEEE QPAIN*, 2025. DOI: <https://doi.org/10.1109/QPAIN66474.2025.11171711>
- J. Hassan, Md Robiul Islam, et al., “EEG-Based Workload Estimation and Classification Using Discrete Wavelet Transform and Statistical Features,” *IEEE QPAIN*, 2025. DOI: <https://doi.org/10.1109/QPAIN66474.2025.11171964>
- Md Khan, J. Hasan, Md Robiul Islam, et al., “Optimizing Pomegranate Disease Detection: A Comparative Analysis of Deep Learning Models,” *IEEE QPAIN*, 2025. DOI: <https://doi.org/10.1109/QPAIN66474.2025.11171672>
- Md. O. Khan, J. Hasan, Md Robiul Islam, et al., “Design, Structural Analysis, and Challenges of High-Voltage Underground Cables in Sindh, Pakistan,” *IEEE SPICSCON*, Nov. 2025. DOI: <https://doi.org/10.1109/SPICSCON69221.2025.11504100>
- Md. O. Khan, J. Hasan, Md Robiul Islam, et al., “A Wavelet-Statistical Method for EEG Signal Classification of Mental Stress in Cognitive Tasks,” *IEEE SPICSCON*, Nov. 2025. DOI: <https://doi.org/10.1109/SPICSCON69221.2025.11503956>
- Md Safiullah, Md Robiul Islam, et al., “Comparative Evaluation of CNN and Transformer Architectures for Multi-Class Lung Disease Classification Using Chest X-Rays,” *IEEE ICCIT*, Dec. 2025. DOI: <https://doi.org/10.1109/ICCIT68739.2025.11490441>

TECHNICAL SKILLS

- **Programming and Simulation:** Python, MATLAB/Simulink, C, C++, MATLAB Communications Toolbox, MATLAB 5G Toolbox, NS-3, GNU Radio, STK
- **Wireless Communications & Networking:** 5G/6G Wireless Networks, 5G NR, Non-Terrestrial Networks (NTN), Satellite Communications, Satellite-Terrestrial Integrated Networks, RF Communication Systems, Digital Modulation Techniques, OFDM, MIMO/Massive MIMO, Beamforming, Wireless Channel Modeling, Spectrum Management, Resource Allocation, Network Optimization, IoT Communications
- **Artificial Intelligence for Wireless Systems:** Deep Reinforcement Learning (DRL), Multi-Agent Reinforcement Learning (MARL), Reinforcement Learning-based Network Optimization, Intelligent Resource Allocation, Autonomous Network Control, AI-enabled Communication Systems
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Neural Networks (ANN), CNN, RNN, LSTM, GANs, Autoencoders, Graph Neural Networks (GNN), Transformers, Attention Mechanisms
- **Reinforcement Learning Frameworks:** PPO, DDPG, SAC, DQN, TD3, Q-Learning, Policy Gradient Methods, Stable-Baselines3, OpenAI Gym
- **Optimization:** Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Simulated Annealing (SA)
- **Cloud & DevOps:** AWS, GCP, Microsoft Azure, Git/GitHub, High-Performance GPU Computing
- **Visualization & BI:** Matplotlib, Seaborn, Plotly, Power BI, Tableau, Google Data Studio

KEY PROJECTS

- MCP Server: Secure Middleware for LLM Integration

2025

 - Built secure Model Context Protocol (MCP) server enabling standardized LLM–enterprise communication with OAuth 2.0 authentication, RBAC, TLS/SSL encryption, and real-time REST/WebSocket integrations for SQL databases and collaboration tools
- Cyber-Physical Systems: Car Dynamics Simulation with Road Grade Disturbance

2025

 - Simulated vehicle dynamics on flat and graded roads (4.5° and 9°) using Newton's second law and the Trapezoidal numerical method; visualized position and velocity outputs to inform vehicle control system design
- Image Restoration Using Spatial and Frequency Domain Filtering

2024

 - Analyzed degradation types (Salt & Pepper, Gaussian, Speckle, Periodic Noise) and applied Median and Wiener filters across spatial and frequency domains to restore image quality and minimize noise artifacts

PROFESSIONAL EXPERIENCE

- Power System Engineer Intern

Nov 2018 – May 2019

Raj Lanka 52.2 MW Power Company Limited, BD

 - Tested electrical components for resistance defects; supported design and construction of power plant and transmission systems
 - Collaborated with senior engineers to troubleshoot and validate new electrical products and distribution infrastructure

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification — Stanford University / Andrew Ng (Coursera)
- Unsupervised Learning, Recommenders, Reinforcement Learning — Stanford University / Andrew Ng (Coursera)
- Advanced Learning Algorithms — Stanford University / Andrew Ng (Coursera)
- Python for Data Science, AI & Development — IBM (Coursera)
- Data Structures and Algorithms Specialization — University of California San Diego (Coursera)

PROFESSIONAL AFFILIATIONS, LEADERSHIP & AWARDS

- Associate Member, IEEE — Pabna University of Science and Technology Student Branch

2018–2020
- Secretary, South Asian Student Association (SASA), Tennessee State University

2024–2025
- Math Olympiad Runner-up — District Level

2011
- Prime Bank Foundation Meritorious Scholarship — Higher Secondary School Certificate

2013–2019
- Dutch-Bangla Bank Scholarship — Secondary School Certificate Level

2010–2012
- Junior School Government Scholarship — Talent Pool Category

2008–2010

REFERENCES

- Dr. Hayder Al-Hraishawi:** Assistant Professor, Dept. of ECE, USF | hayder@usf.edu | +1 813-974-2999
- Dr. Sagnika Ghosh:** Associate Professor, Dept. of ECE, TSU | sghosh1@tnstate.edu | +1 615-963-5362
- Dr. Kamrul Hasan:** Assistant Professor, Dept. of ECE, TSU | mhasan1@tnstate.edu | +1 615-963-7539
- Dr. Md Feroz Ali:** Associate Professor, Dept. of EEE, PUST | feroz.ali.eee@pust.ac.bd | +8801724909432